

RC Circuit Response to Step, Pulse, and Square Inputs

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1 Step Input

Step signal is represented as:

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$$

Circuit:

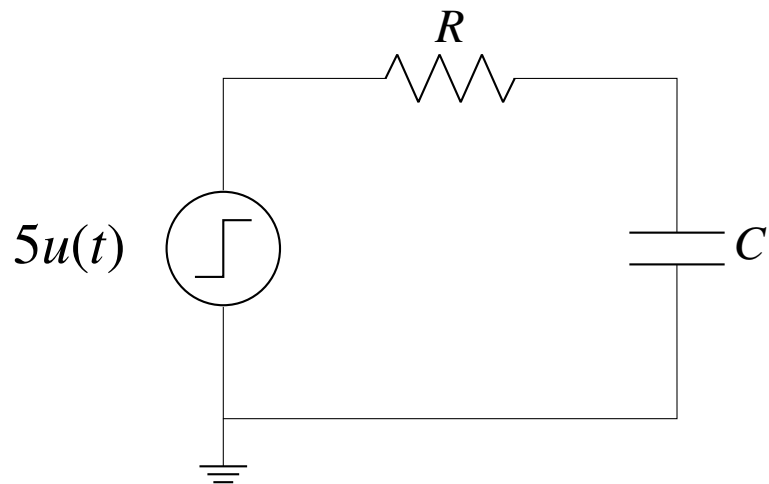


Figure 1: RC Circuit with Step Input

Charging waveform: The sum of the voltage drops across the resistor (v_R) and capacitor (v_C) must equal the source voltage:

$$\Rightarrow 5u(t) = v_R(t) + v_C(t) \quad (1)$$

$$\Rightarrow v_R(t) = R \left(C \frac{dv_c}{dt} \right) = RC \frac{dv_c}{dt} \quad (2)$$

$$\Rightarrow 5u(t) = RC \frac{dv_c}{dt} + v_c(t) \quad (3)$$

$$\Rightarrow 5u(t) - v_c(t) = RC \frac{dv_c}{dt} \quad (4)$$

$$\Rightarrow \frac{dt}{RC} = \frac{dv_c}{5u(t) - v_c(t)} \quad (5)$$

Integrating both sides from $t = 0$ to t . We assume the capacitor is initially uncharged ($v_c(0) = 0$)

$$\int_0^t \frac{1}{RC} dt = \int_0^{v_c(t)} \frac{1}{5u(t) - v} dv \quad (6)$$

$$\frac{t}{RC} = [-\ln(5u(t) - v)]_0^{v_c(t)} \quad (7)$$

$$-\frac{t}{RC} = \ln(5u(t) - v_c(t)) - \ln(5u(t) - 0) \quad (8)$$

$$-\frac{t}{RC} = \ln \left(\frac{5u(t) - v_c(t)}{5u(t)} \right) \quad (9)$$

$$\Rightarrow e^{-t/RC} = \frac{5u(t) - v_c(t)}{5u(t)} \quad (10)$$

$$\Rightarrow 5u(t)e^{-t/RC} = 5u(t) - v_c(t) \quad (11)$$

$$\boxed{v_c(t) = 5u(t) \left(1 - e^{-\frac{t}{RC}} \right)} \quad (12)$$

For our problem the voltage of capacitor at time t is given by:

$$v_c(t) = 5 \left(1 - e^{-\frac{t}{RC}} \right) \quad (13)$$

similarly for discharging waveform we get:

$$v_c(t) = 5e^{-\frac{t}{RC}} \quad (14)$$

Inference

- From (13) and (14) we can say that the voltage of the capacitor increases exponentially in the case of charging and decreases exponentially in the case of discharging.
- The same is also observed in the below waveforms.

1.1 Waveforms

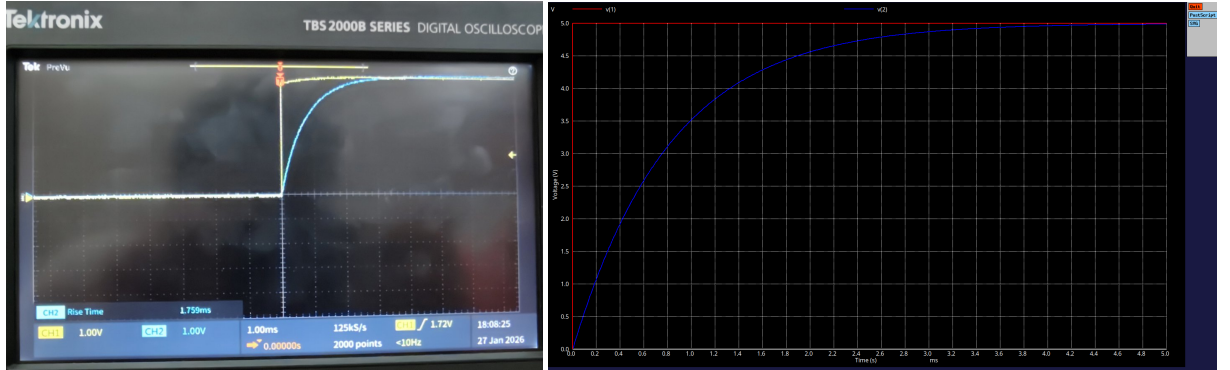


Figure 2: Charging Waveform verified with ngspice simulation

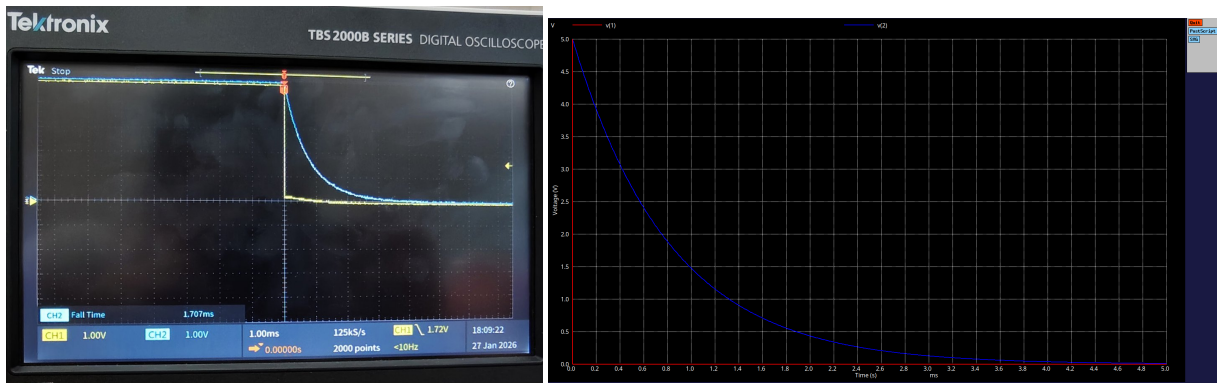


Figure 3: Discharging Waveform verified with ngspice simulation

1.2 Time constant calculation

$$v_c(t) = 5 \left(1 - e^{-\frac{t}{RC}} \right) \quad (15)$$

At $t = RC$

$$v_c(t) = 5 \left(1 - e^{-1} \right) \quad (16)$$

$$\Rightarrow v_c(t) = 5(1 - 0.632) = 3.16V \quad (17)$$

i.e time to taken raise the voltage of capacitor from 0 to 63.2% of it's maximum voltage is defined as Time constant(τ) of the circuit.

From the waveforms obtained in DSO we get the time constant as

$$\tau \approx 820\mu s \quad (18)$$

Rise Time:

The rise time (t_r) of an RC circuit is defined as the time required for the voltage across the capacitor to rise from 10% to 90% of its final steady-state value.

$$\Rightarrow \tau = \frac{t_r}{2.2} \quad (19)$$

$$\Rightarrow \tau = \frac{1759}{2.2} \mu s \quad (20)$$

$$\tau \approx 800 \mu s \quad (21)$$

Error:

$$\text{Error} = \frac{820 - 800}{820} \times 100\% \quad (22)$$

$$\Rightarrow = \frac{20}{820} \times 100\% \quad (23)$$

$$\Rightarrow \text{Error} \approx 2.43\% \quad (24)$$

Experimental value of time constant almost matches the original value of time constant with an error $<2.5\%$

Conclusion:

- In the case of a charging waveform for step input for $t > 0$ the capacitor will charge exponentially according to the relation $v_c(t) = 5 \left(1 - e^{-\frac{t}{RC}}\right)$
- In the case of a discharging waveform for a step input the charged capacitor will start discharging exponentially according to the relation $v_c(t) = 5 \left(e^{-\frac{t}{RC}}\right)$
- The above results match exactly with the ngspice simulations.

2 Pulse Input

Pulse input is given by:

$$x(t) = \begin{cases} 0, & t < 0 \\ 5, & 0 \leq t < T \\ 0, & t \geq T \end{cases}$$

Circuit:

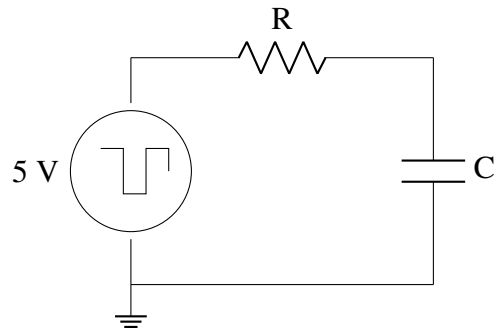


Figure 4: RC Circuit with Pulse Input

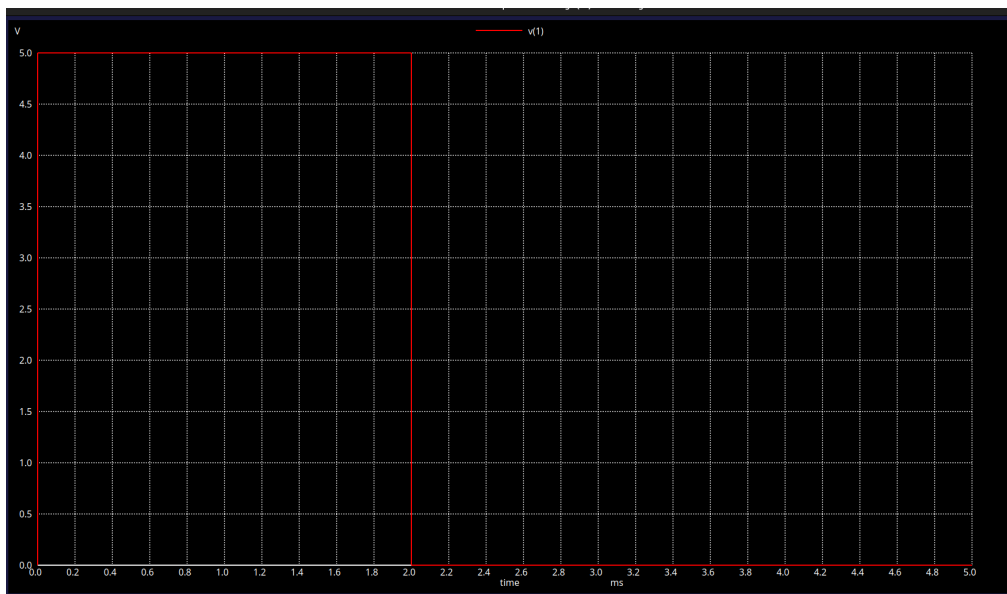


Figure 5: Single Pulse Input

Equation

$$v_c(t) = 5 \left(1 - e^{-\frac{t}{RC}} \right) \quad (25)$$

Explanation:

- The peak of the capacitor voltage depends on the duty cycle of the wave.
- If the duty cycle is low then the capacitor will not get enough time to charge completely hence the peak voltage of the capacitor in this case will be less than the 5v
- If the duty cycle is high enough then the capacitor will get enough time to charge and the peak voltage will be $\approx 5v$

- After the time period of the wave the capacitor will start discharging and voltage of the capacitor decreases to 0 exponentially.

2.1 Wave forms

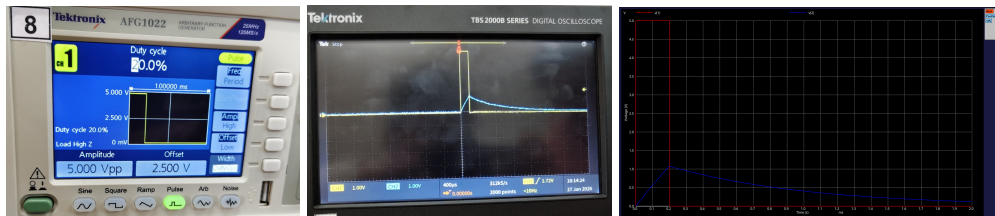


Figure 6: RC Circuit with Pulse of with 20% duty cycle

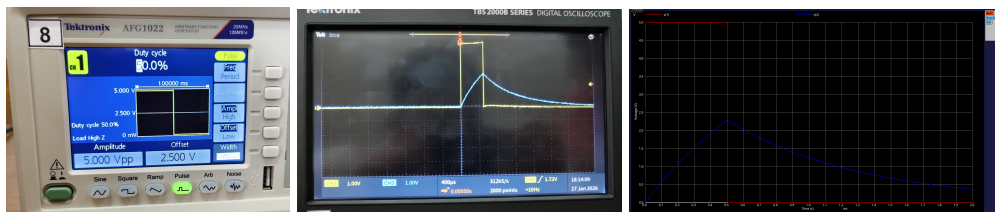


Figure 7: RC Circuit with Pulse Input of 50% duty cycle

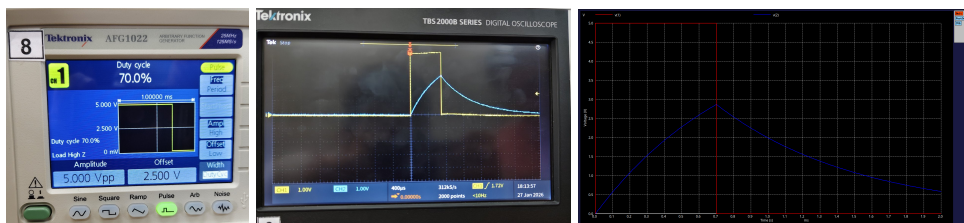


Figure 8: RC Circuit with Pulse Input of 70% duty cycle

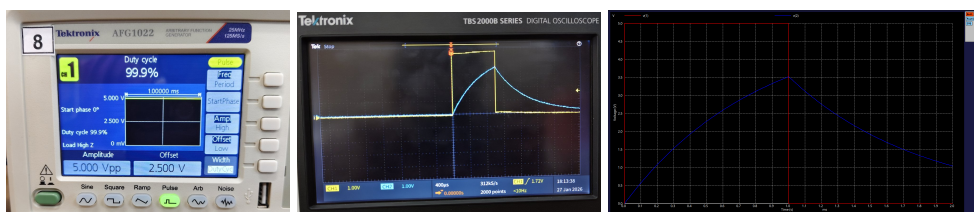


Figure 9: RC Circuit with Pulse Input of 99.9% duty cycle

2.2 Calculations

20% duty cycle:

$$R = 820 \Omega \quad (26)$$

$$t_{\text{on}} = 1 \text{ ms} \times 0.20 \quad (27)$$

$$t_{\text{on}} = 200 \times 10^{-6} \text{ s} \quad (28)$$

$$\tau = 820 \mu\text{s} \quad (29)$$

$$V_{\text{peak}} = 5 \left(1 - e^{-\frac{t_{\text{on}}}{RC}} \right) \quad (30)$$

$$\Rightarrow V_{\text{peak}} = 5 \left(1 - e^{-0.243} \right) \quad (31)$$

$$\Rightarrow V_{\text{peak}} \approx \mathbf{1.08 \text{ V}} \quad (32)$$

Similarly on varying duty cycle we get different values of V_{peak}

Comparison Table :

Duty Cycle(%)	$t_{\text{on}}(\mu\text{s})$	$V_{\text{peak}}(\text{V})$
20.0	200	1.08V
50.0	500	2.28V
70.0	800	2.87V
99.9	999	3.52V

Conclusion

- As the duty cycle increases the peak voltage of the capacitor also increases.
- Maximum on time = $1000 \mu\text{s}$ i.e $\approx 1.22\tau$
- Capacitor takes $\approx 5\tau$ to charge completely but here maximum on time is less than 5τ .so, capacitor can't charge completely in this time
- capacitor charges to peak voltage $< 5\text{V}$ in each case depending on the duty cycle and discharges after that time exponentially.
- The same is observed experimentally and is also verified by ngspice simulations

3 Square-Wave input

The square wave input voltage $v_{\text{in}}(t)$ with a period T is defined as:

$$v_{\text{in}}(t) = \begin{cases} 5, & 0 \leq t < \frac{T}{2} \\ 0, & \frac{T}{2} \leq t < T \end{cases} \quad (33)$$

with the periodicity condition $v_{in}(t + T) = v_{in}(t)$.

Circuit:

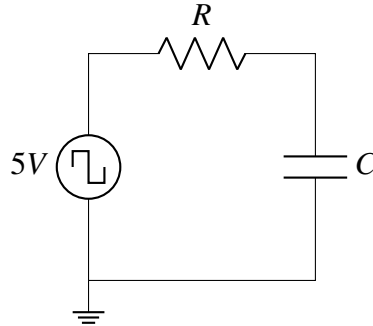


Figure 10: RC Circuit with a 5V Square Wave Input

3.1 $RC \gg T$

In this case the value of $\tau = 820\mu s$ so, let's take the frequency of the square wave as 100kHz. Then, $T = 10\mu s$

Calculation:

- $v_c(t) = 5 \left(1 - e^{-\frac{t}{\tau}}\right)$ where $\tau = 820\mu s$.
- Capacitor takes ≈ 5 time constants to charge. $\tau \approx 4.1ms$.
- The pulse width is very short ($5\mu s$). This is less than 1% of the time constant. During the "ON" cycle, the capacitor has time to begin charging before the input switches "OFF".
- The capacitor cannot follow the rapid switching of the input. It settles at the point where the charge accumulating during the "ON" phase $\approx$ equals the charge draining during the "OFF" phase.
- The graph looks like a straight line at 2.5V. But in real the graph oscillates very less at 2.5V.

For the above reason we can't capture the transient response of the capacitor in this case.

Waveforms

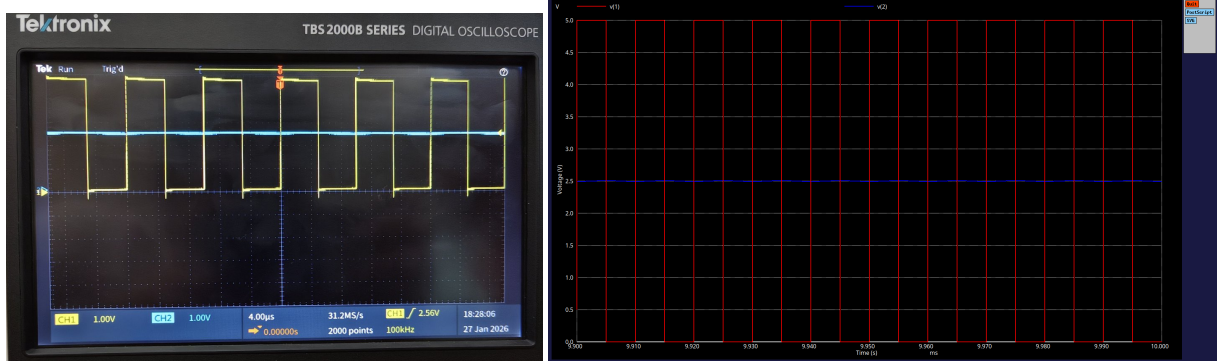


Figure 11: $RC \ll T$ case verified with ngspice simulation

3.2 $RC \approx T$

The value of $\tau = 820\mu s$ so, let's take the frequency of the square wave as 1kHz. Then, $T = 1000\mu s$.

For a square wave of time period T its value is 5V for $\frac{T}{2}$ and 0 for remaining time.

Calculation:

The voltage across the capacitor $v_c(t)$ for a square wave input is defined as:

$$v_c(t) = \begin{cases} 5 + (V_{min} - 5)e^{-\frac{t}{\tau}} & \text{for } 0 \leq t < \frac{T}{2} \quad (\text{Charging}) \\ V_{max} \cdot e^{-\frac{t - \frac{T}{2}}{\tau}} & \text{for } \frac{T}{2} \leq t < T \quad (\text{Discharging}) \end{cases} \quad (34)$$

V_{min} and V_{max} can be found by applying the following boundary conditions

$$v_c\left(\frac{T}{2}\right) = V_{max} = 5 + (V_{min} - 5)e^{-\frac{T}{2\tau}} \quad (35)$$

$$v_c(T) = V_{min} = V_{max} e^{-\frac{T}{2\tau}} \quad (36)$$

on solving (35) and (36) we get

$$V_{min} = 1.76V \quad (37)$$

$$V_{max} = 3.24V \quad (38)$$

In this case we can clearly capture transient and steady state responses.

Wave forms

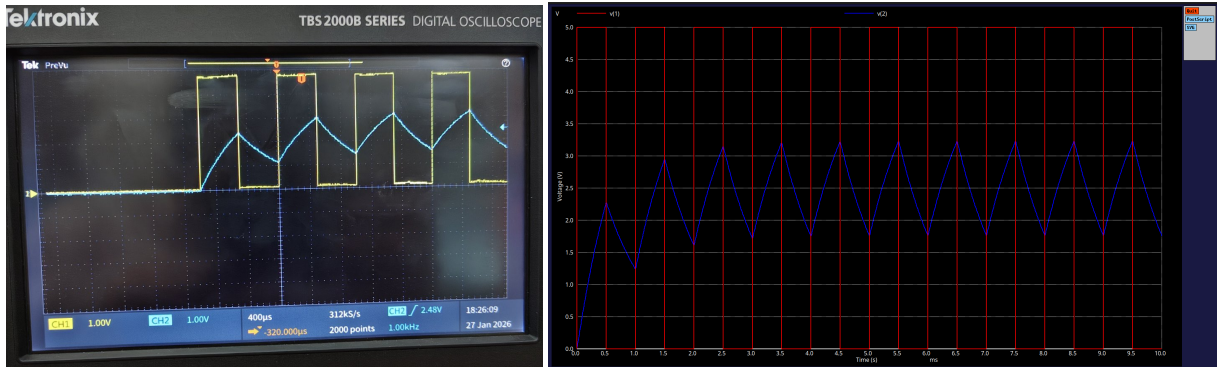


Figure 12: $RC \approx T$ case Transient response verified with ngspice simulation

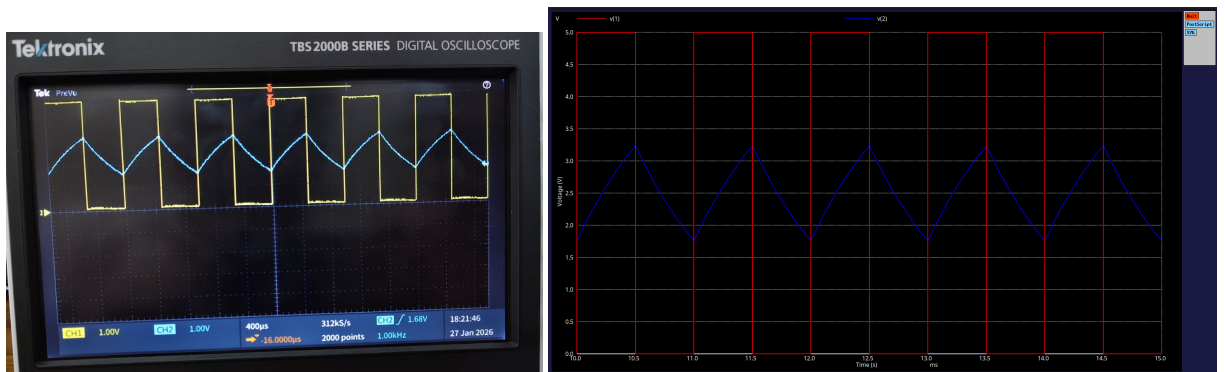


Figure 13: $RC \approx T$ case Steady state verified with ngspice simulation

So, in steady state capacitor oscillates between voltage of 1.76V and 3.24V with an average of 2.5V.

3.3 $RC \ll T$

In this case the value of $\tau = 820\mu s$ so, let's take the frequency of the square wave as 1Hz. Then, $T = 1s$.

Calculation:

- $v_c(t) = 5 \left(1 - e^{-\frac{t}{\tau}}\right)$ where $\tau = 820\mu s$.
- Capacitor takes ≈ 5 time constants to charge, i.e. $\approx 4.1ms$. On time of the capacitor is 500ms, so, in $< 1\%$ of total on time only the capacitor gets charged completely.

- The output waveform appears identical to the input square wave. It switches fully between 0V and 5V . The characteristic exponential curve is not visible to the naked eye at this time scale.
- The charging happens in less than 1% of the total pulse width. So, the transient response is negligible compared to the signal period. we can't capture transient response in this case.

Waveforms:

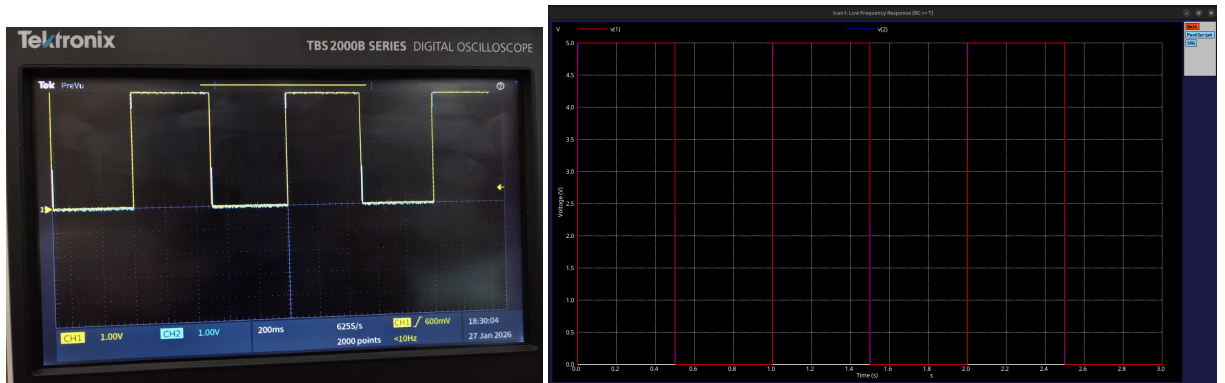


Figure 14: $RC \ll T$ case Steady state verified with ngspice simulation

As discussed above the output appears identical to input and overlaps with input due to which we can see pink color lines in the ngspice simulation.

Comparison Table:

Feature	Low Frequency	Medium Frequency	High Frequency
Condition	$RC \ll T$	$RC \approx T$	$RC \gg T$
Output Shape	Square Wave (Identical to Input)	Triangular / Exponential	Flat DC Line (Constant Voltage)
Capacitor Status	Fully charges and discharges in every cycle.	Partially charges and discharges.	Barely charges before switching; holds average charge.
Output Voltage	Full Swing (0V $\leftrightarrow$ 5V)	Partial Swing (1.76V $\leftrightarrow$ 3.24V)	Steady DC ($\approx$ 2.5V)

Table 1: Comparison of RC Circuit Response across three different frequencies ($R = 820\Omega$, $C = 1\mu F$).